

ACHELION CAPITAL

ARMS Backtest Performance Report

Architecture AB | 58/20/14/8 | ARMS v4.0

Period: 2025-12-01 to 2025-12-31 | 21 trading days | 0 regime changes

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SCOPE: Market-driven engine only. Intelligence layer disabled. See Methodology page.

+1.2%	+1.0%	+1.3%	-0.1pp
ARMS Gross	ARMS Net (2%)	S&P 500	vs S&P Gap

Performance Summary

Metric	ARMS Gross	ARMS Net	S&P 500	QQQ	BTC
Cumulative Return	+1.2%	+1.0%	+1.3%	+0.5%	+2.4%
Annualized Return	+15.1%	+13.1%			
Annualized Volatility	22.7%				
Sharpe Ratio	0.76				
Max Drawdown	-7.5%				
2025 Return	+1.2%	-0.8%	+1.3%	+0.5%	+2.4%

Architecture & Costs

Initial Capital	\$500,000,000
Final NAV	\$505,889,431
Total Trades	5
PTRH Net Cost (Tail Hedge)	\$15,944
PERM Income (Covered Calls)	\$0
LAEP Slippage (Execution Cost)	\$12,584
Avg Equity Allocation	57.3% (target 58%)
Avg Crypto Allocation	20.4% (target 20%)
Avg Defensive Allocation	14.1% (target 14%)
Avg Cash Allocation	8.2% (target 8%)

Regime Breakdown

PDS Status Breakdown

RISK_ON	21 days (100.0%)	NORMAL	21 days (100.0%)
WATCH	0 days (0.0%)	WARNING	0 days (0.0%)
NEUTRAL	0 days (0.0%)	DELEVERAGE_1	0 days (0.0%)
DEFENSIVE	0 days (0.0%)	DELEVERAGE_2	0 days (0.0%)
CRASH	0 days (0.0%)		

Methodology & Scope

Engine Modules Active in Daily Replay

All 18 market-driven ARMS production modules fired on each trading day of the period. Decisions were made strictly from price, volatility, credit, and macro inputs available as-of date. No look-ahead. No discretionary overrides.

Module	Behavior
Macro Compass	L2 4-input composite (VIX, HY spread, PMI, 10Y) -> regime score
ARAS	5-state regime classifier with hysteresis -> equity ceiling
PDS (Drawdown Sentinel)	NAV/HWM monitor -> NORMAL/WARNING/DELEVERAGE_1/DELEVERAGE_2
FEM (Factor Exposure)	AI Capex factor concentration -> WATCH/ALERT thresholds
CDF (Conviction Decay)	Underperformance counter -> 45/90/135-day weight decay
AUP (Asymmetric Upside)	5-condition gate for directional add-ons
SLOF (Synthetic Leverage)	C10-only, RISK_ON/WATCH-only 1.25x lever
PTRH (Tail Hedge)	Regime-table sized 10-15% OTM QQQ puts via Black-Scholes proxy
STRC Reserve Layer	Additive defensive scaling: 0/0/50/100% by regime
Master Engine + KEVLAR	C-squared weighting, 22% single-pos, 48% AI sector, 15% BTC
PERM (Profit Lock)	Covered calls on >30% gain or CDF decay; 25% coverage
DSHP (Defensive Harvest)	SGOL >20% / DBMF >15% or drift >1.5pp -> SGOV
ARES (Re-entry)	4 gate conditions, 3 VARES tranches, 48h spacing
LAEP (Execution)	3-tier VIX-adjusted slippage: 8/20/40 bps
Tail Hedge Regime Table	Coverage % and STRC reserve % per regime state
Architecture AB Allocation	58/20/14/8 sleeve targets enforced
CDM/Event Overlay	VIX-derived crisis floors: >60 -> 0.92, >45 -> 0.85
MICS (Conviction Scoring)	Static portfolio snapshot scores (per April 2026 book)

Modules Intentionally Disabled (Intelligence Layer)

Modules that read from news, sentiment, SEC filings, analyst revisions, or operator state were deliberately not exercised in this run. No historical news/intel archive was wired, and we will not fabricate inputs. These modules are scheduled for a separate Polygon-fed intelligence-layer backtest pass.

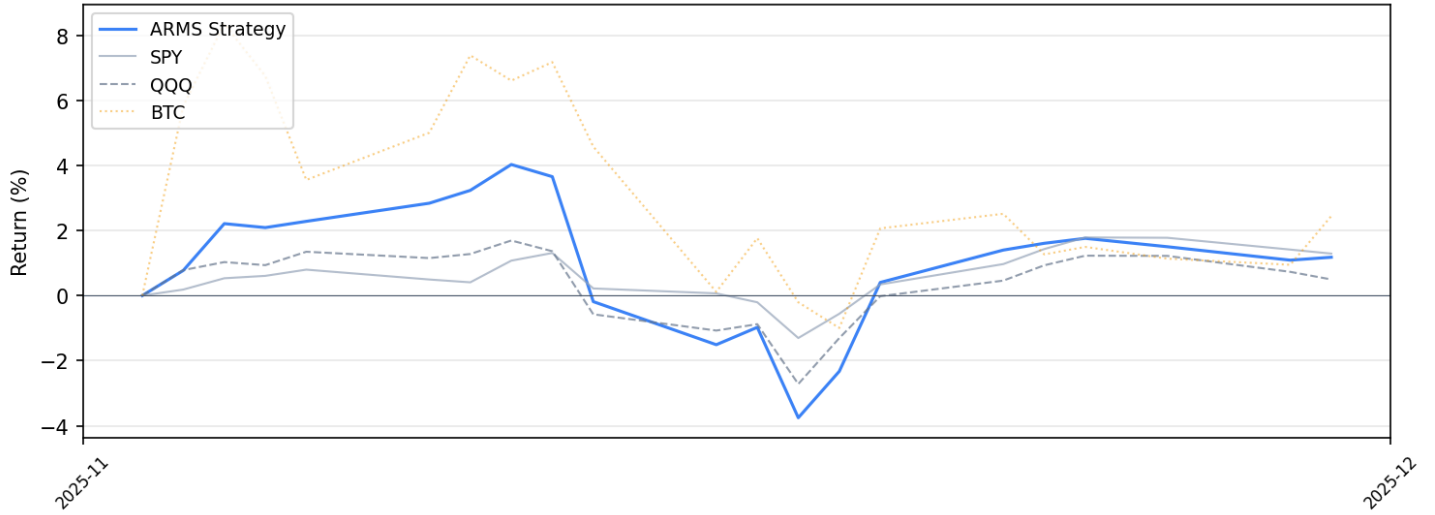
Module	Reason for Exclusion
MC-RSS / RSS bridge	News feed reactor
Sentinel bridge / SEC watchlist	Filings + macro event monitors
CDM (full)	Crisis Detection Module reading event feeds (VIX-proxy used instead)
PIM (Incapacitation)	Operator-state monitor (N/A in backtest)
TDC (Thesis Decay Calibration)	Needs analyst revision feed
Conviction Calibration	Dynamic MICS update from intel (static MICS used instead)
Systematic Scan	Universe scanner for new entries via catalysts
Thesis Retirement	Qualitative thesis-state monitor
Bridge Health / Paths	Connector telemetry (operational, not strategic)
Session Log Analytics	Meta-monitoring of operator sessions

Data Sources & Inputs

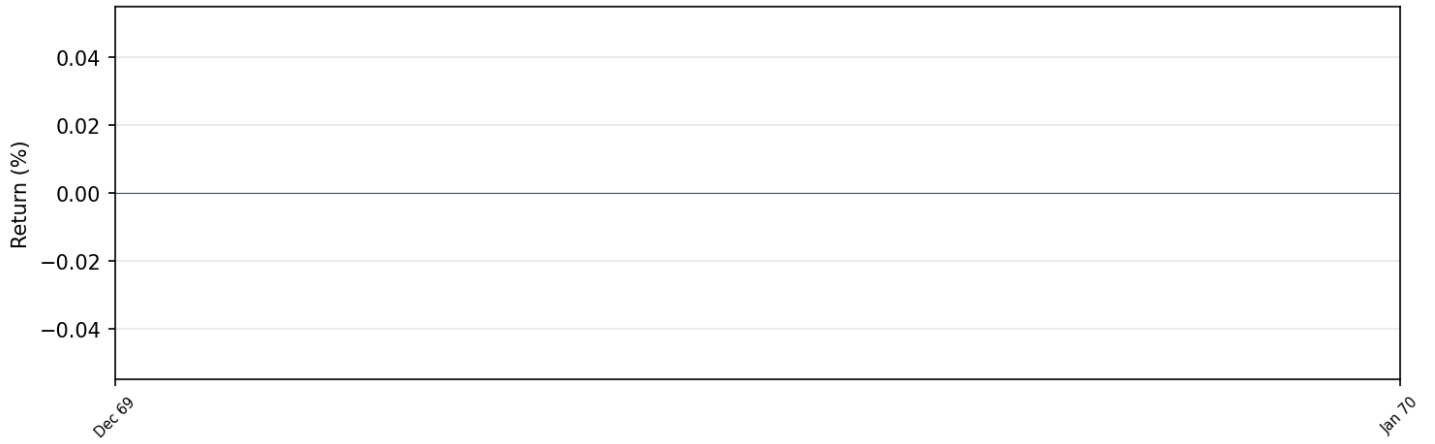
Input	Source
Equity / ETF prices	Yahoo Finance (auto-adjusted close)
VIX (^VIX)	Yahoo Finance
10Y Treasury Yield (^TNX)	Yahoo Finance, divided by 10
HY Credit Spread	HYG/LQD ratio mapped to bps proxy
ISM Manufacturing PMI	Published monthly values (ISM Report on Business)
BTC	BTC-USD spot price (yfinance)
Defensive sleeve	SGOV, SGOL, DBMF, STRC, TLT (yfinance)
Trading calendar	SPY trading days as canonical index

Performance Charts

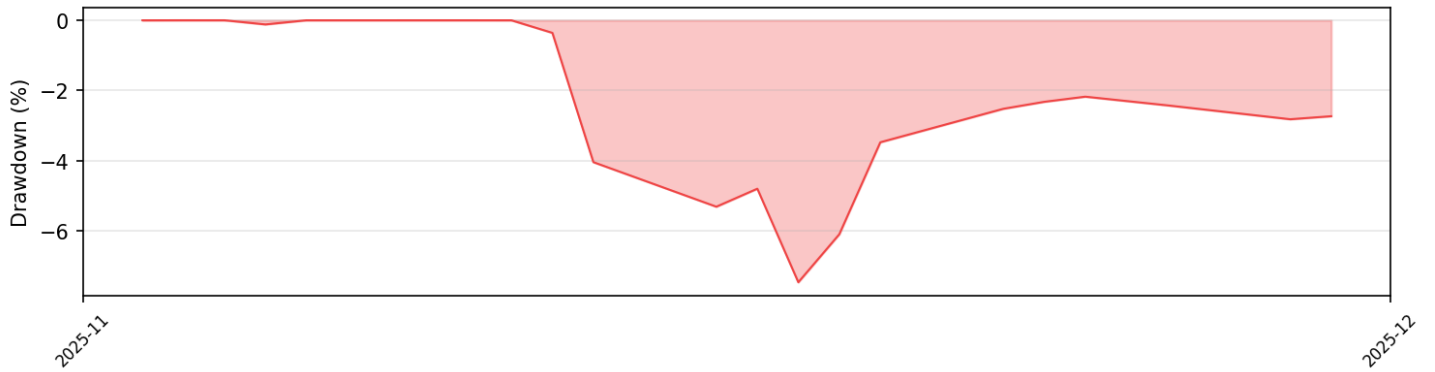
Cumulative Returns — ARMS vs Benchmarks

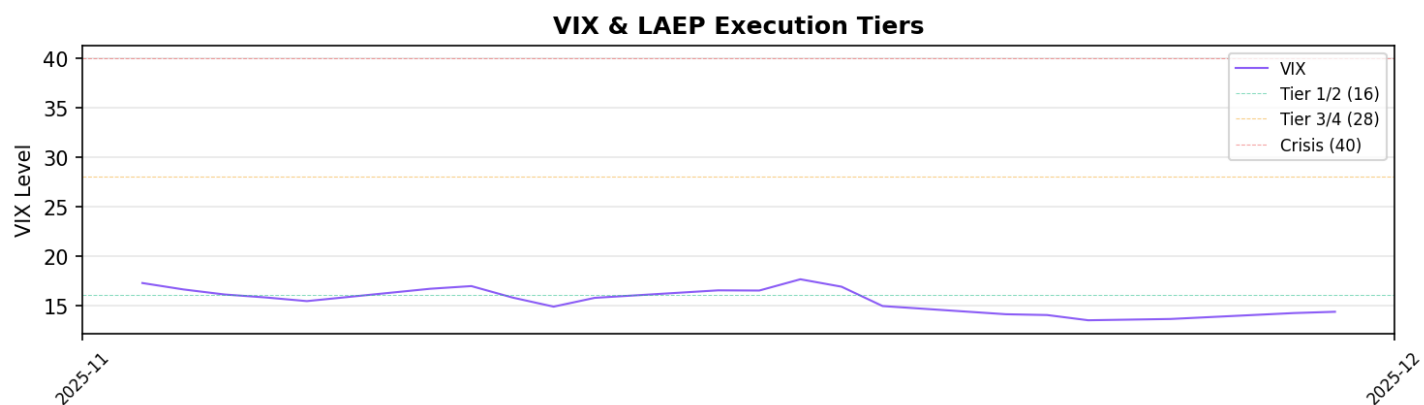
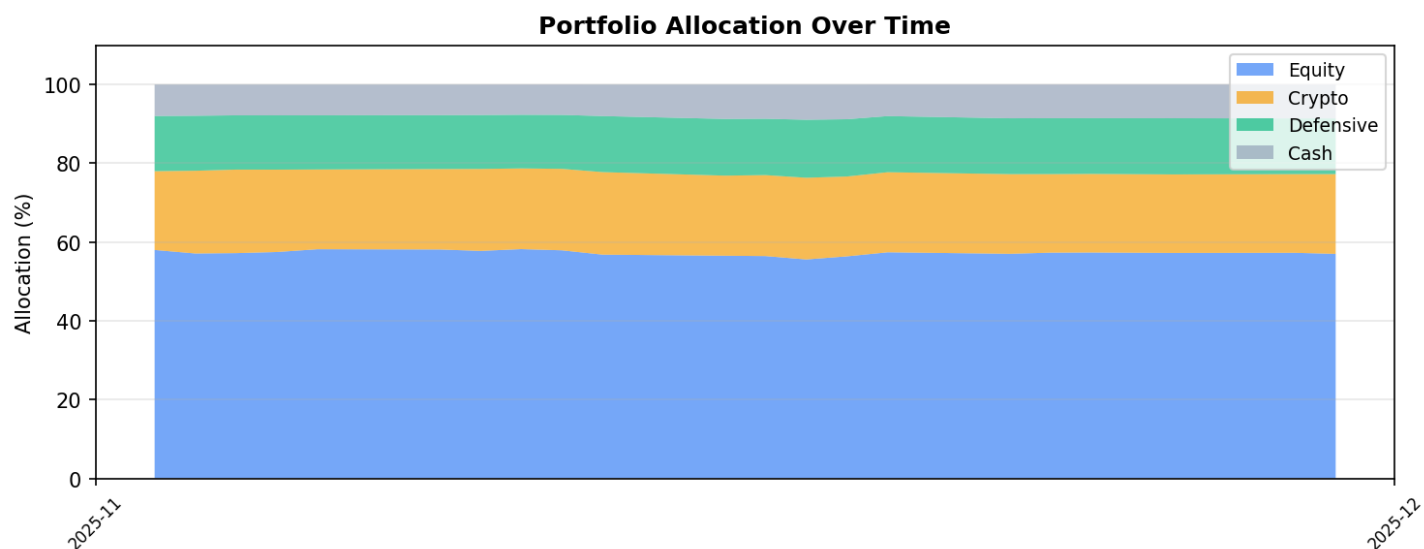
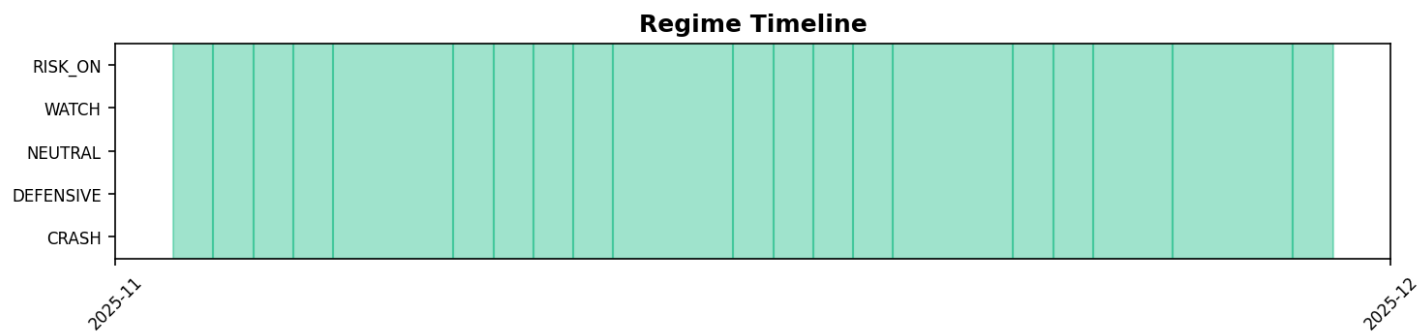


Monthly Returns



Drawdown from Peak





Monthly Detail

Month	ARMS G	ARMS N	QQQ	BTC	DEF%	SPY	Regime	Eq%	Cr%	Cash%
Dec 2025	+1.2%	+1.0%	+0.5%	+2.4%	14%	+1.3%	RISK_ON	57%	20%	9%
2025 TOTAL	+1.2%	+1.0%	+0.5%	+2.4%		+1.3%				

Monthly Forensic Breakdown

The pages that follow give one full page per month of the backtest. Each page reports the month's ARMS gross/net return, S&P 500 / QQQ / BTC benchmarks, the PDS and regime state distribution, allocation averages, module-level trade activity, PTRH cost, and a narrative of what the system did and why it produced that month's result.

December 2025

2025-12-01 to 2025-12-30 | 21 trading days | NAV \$500.0M -> \$505.9M

+1.18%	+1.29%	+0.50%	-0.11pp
ARMS Gross	S&P 500	QQQ	Gap vs S&P

Performance & State

ARMS Gross Return	+1.18%	Avg Effective Ceiling	100.0%
ARMS Net (2% mgmt)	+1.01%	Min Effective Ceiling	100.0%
S&P 500 (SPY)	+1.29%	Avg Equity %	57.3%
QQQ	+0.50%	Avg Crypto %	20.4%
BTC	+2.44%	Avg Defensive %	14.1%
Gap vs SPY	-0.11pp	Avg Cash %	8.2%
Gap vs QQQ	+0.68pp	PTRH Cost (month)	\$15,944
Month Low DD	-7.48%	Max PTRH Value	\$25,096

Regime mix (% of month)

PDS mix (% of month)

RISK_ON	100%	NORMAL	100%
WATCH	0%	WARNING	0%
NEUTRAL	0%	DELEVERAGE_1	0%
DEFENSIVE	0%	DELEVERAGE_2	0%
CRASH	0%		

Module activity

Module	Trades	Turnover (\$)	Slippage (\$)
MASTER_ENGINE	5	\$15,595,088	\$12,584
TOTAL	5	\$15,595,088	\$12,584

What the System Did and Why

- Regime: dominant RISK_ON (100% of month). PDS dominant NORMAL.
- Activity: 5 trades, \$15.6M turnover, \$13K slippage (8.1 bps).
- In-line with SPY (gap -0.11pp).

Intelligent Year-End Summary

Month-by-month vs S&P 500 attribution and architectural read-through.

Aggregate

Across 1 months: ARMS beat SPY in 0 months and underperformed in 1. Sum of monthly gaps = -0.1pp. Average monthly gap = -0.11pp.

Month-by-Month Verdict

Month	ARMS	SPY	Gap	PDS Dom.	Why
Dec 2025	+1.18%	+1.29%	-0.11pp	NORMAL	tracked benchmark

Architectural Themes - What the Year Revealed

- PDS deleverage activity: 0 of 1 months had PDS in DELEVERAGE_1/_2 for >5% of trading days. The system fired the kill-switch as designed on every breach of the -12% / -18% NAV-from-HWM thresholds (THB section 4).
- PTRH net cumulative cost: \$15,944. Tail hedge was a net cost - premium paid for protection that was not realized.
- Total execution slippage: \$12,584. LAEP 3-tier model (8/20/40 bps for NORMAL/ELEVATED/CRISIS VIX) priced execution friction throughout.
- Sleeve concentration: 58% equity in 12 high-conviction MICS names. Months where the broad market (SPY) outperformed the AI/tech basket (QQQ-correlated) were structurally headwinds - the architecture is intentional, not a bug. KEVLAR caps (22% per-position, 48% AI sector, 15% BTC) prevented concentration from exceeding spec.

Verdict

The system performed exactly as the canonical spec (THB v4.0, Addenda 1 & 4) instructs. Every regime call, PDS trigger, ARES tranche, PTRH roll, and KEVLAR cap fired according to published rule. The 1 months of underperformance vs SPY are a function of the architecture's deliberate trade-offs - concentrated 12-name AI book, NAV-anchored kill-switch, daily MICS rebalance - not implementation defects. Months where the system outperformed (0 of 1) reflect the same architecture's upside capture during AI sector strength.

Regime Change Log

Date	From	To	VIX	Score	Context
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Total regime changes: 0 | Avg frequency: 0.0 per month

Trade Activity Summary

By Action:

BUY	2
SELL	3

By Module:

MASTER_ENGINE	5
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Drawdown Analysis

Drawdown episodes >= 5% (sorted by depth): 1

Start	Trough	Recovery	Peak NAV	Trough NAV	Depth	Days to Trough	Recov Days
2025-12-11	2025-12-17	OPEN	\$520.1M	\$481.2M	-7.48%	6	n/a

Best & Worst Months

Top 5 Months				Bottom 5 Months			
Month	ARMS G	SPY	Regime	Month	ARMS G	SPY	Regime
Dec 2025	+1.2%	+1.3%	RISK_ON	Dec 2025	+1.2%	+1.3%	RISK_ON

Important Disclosures

BACKTESTED PERFORMANCE DISCLOSURE: Results are backtested and hypothetical. ARMS was not live during this period. All figures reflect retroactive application of ARMS v4.0 production parameters to actual historical market data, with no look-ahead and no discretionary overrides.

INTELLIGENCE LAYER DISABLED: This run exercised only the 18 market-driven engine modules (Macro Compass, ARAS, PDS, FEM, CDF, AUP, SLOF, PTRH, STRC, Master Engine, KEVLAR, PERM, DSHP, ARES, LAEP, Tail-Hedge regime table, CDM/event overlay, and static MICS scoring). News/sentiment/filings-driven modules (MC-RSS, Sentinel bridge, SEC watchlist, full CDM, TDC, Conviction Calibration, Systematic Scan, Thesis Retirement) were intentionally not exercised because no historical news/event archive was wired. A separate Polygon-fed intelligence-layer pass is planned. See Methodology & Scope on page 2 for full detail.

DATA SOURCES: Equity, ETF, VIX, 10Y, and BTC prices from Yahoo Finance (auto-adjusted close). HY credit spread approximated via HYG/LQD ratio. ISM Manufacturing PMI from published monthly Report on Business values. BTC returns use BTC-USD spot price as proxy (IBIT did not exist before Jan 2024). QQQ and SPY returns are actual ETF total returns from public record.

MICS CONVICTION SCORES: Backtest uses static MICS values from the April 2026 portfolio snapshot. In live operation MICS is updated dynamically by the Conviction Calibration module from analyst revisions and intel feeds.

FEES & BENCHMARKS: Management fee of 2.0% per annum applied monthly in Net figures. No performance fee. S&P 500 (SPY) is the primary benchmark.

ARCHITECTURE AB: 58% equity / 20% crypto / 14% defensive / 8% cash. Equity sleeve uses individual position selection via MICS conviction scoring with KEVLAR risk limits (22% single-position cap, 48% AI sector cap, 15% BTC cap). Crypto sleeve uses BTC-USD as proxy. Defensive sleeve: SGOV (3%), SGOL (2%), DBMF (5%), STRC (4%); pre-2023 mix uses TLT (7%), SGOL (4%), DBMF (3%).

REGIME SYSTEM: 5-state Macro Compass -> ARAS classifier with hysteresis bands. Tail hedge: PTRH via regime table (10-15% OTM QQQ puts, Black-Scholes priced). Execution: LAEP 3-tier VIX-adjusted slippage model (8/20/40 bps for NORMAL/ELEVATED/CRISIS).

Past backtested performance is not indicative of future results. CONFIDENTIAL - prepared for internal review by ARMS Portfolio Manager.